

Dheeraj Saraswat

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Education

Vellore Institute of Technology, Bhopal

Bachelor of Technology (Computer Science) – CGPA: 8.08/10

Aug 2023 – Jun 2027

Bhopal, Madhya Pradesh

Technical Skills

Languages: C++, Java, SQL, Python, R

Frameworks and Databases: PostgreSQL, MySQL, Django, Flask, Apache, React.js, Node.js

Platforms: AWS, Salesforce, Power BI, Tableau

Security: Malware Analysis, Threat Intelligence, Incident Response, Reverse Engineering, Network Security

Data Format: JSON

Work Experience

Data Science Intern | Advance Group of Institutions, Delhi

May 26 – Present

- Engineered solar energy forecasting model (Prophet/ARIMA) on multi-variable telemetry (irradiance, humidity, temperature), achieving 8.73% MAPE accuracy across 12-month backtesting.
- Architected scalable ETL pipeline automating sensor data ingestion, schema validation and anomaly imputation, reducing data preparation latency by 70%.
- Delivered real-time Power BI dashboards consolidating energy output, load curves and equipment health KPIs across 3+ plant datasets.

Achievements

- Honoured with the "Best Paper Award" at the FINCON 2025 International Conference for the paper titled "Proactive Fraud Prevention with AI/ML in the Digital Age." The award recognises outstanding research in financial security.
- ACM IKDD Student Travel Grant | CODS 2025 — Awarded a competitive travel grant to attend the premier international conference on Data Science and Management of Data.

Projects

USB Threat Monitor | WMI, python-magic, Windows Internals

Jan 25 – Apr 25

- Built a Python-based USB Threat Monitor using WMI, python-magic, and threading to detect and auto-shutdown BadUSB, Rubber Ducky, and autorun malware within 5 seconds.
- Ensured zero-tolerance data protection in high-security environments through real-time threat response.

AgriScan: Crop Disease Detection | Python, Keras, Flask

Sep 25 – Feb 26

- Designed a 25+ rule epidemiological disease-risk engine (temperature/humidity thresholds per crop) and class-weighting strategy compensating for a 111:1 class imbalance ratio across 57 disease categories.
- Architected a single-cell Colab restoration pipeline reloading a trained MobileNetV2 model, Flask backend, and Cloudflare tunnel in 30 seconds, eliminating retraining after runtime disconnects.
- Resolved a localStorage quota-exceeded production bug by implementing canvas-based image compression (500KB→3KB) and a fallback retry handler, reducing storage footprint by 99% with zero data loss.

Extracurricular Achievements

- Spearheaded a high-profile campus event featuring former ISRO Director Dr. Brinda V, attracting 350+ science-engaged students and fostering a strong commitment to science communication.

Position of Responsibility

Hindi Club, Vice President, VIT Bhopal University

May 24 – May 26

- Led and managed a team of 200+ members, ensuring the smooth execution of club activities.
- Organised multiple cultural events, including Hindi Diwas, Kavyanjali, Hindi Sabha, and a large-scale Holi celebration, engaging over 3,000 students in safe, inclusive, and vibrant campus activities.

Rajasthan Club, Student Coordinator, VIT Bhopal University

May 24 – Jan 25

- Led a 120+ member team, boosting club membership by 150% through new initiatives.
- Orchestrated the Rajasthan Club's inaugural and highly successful event, engaging an audience of 1000+ attendees.